SELECTED WORK

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Machine learning engineer and data scientist

Projects with their data, results and limits.



PROJECT DECK / 2026

MACHINE LEARNING DATA SCIENCE

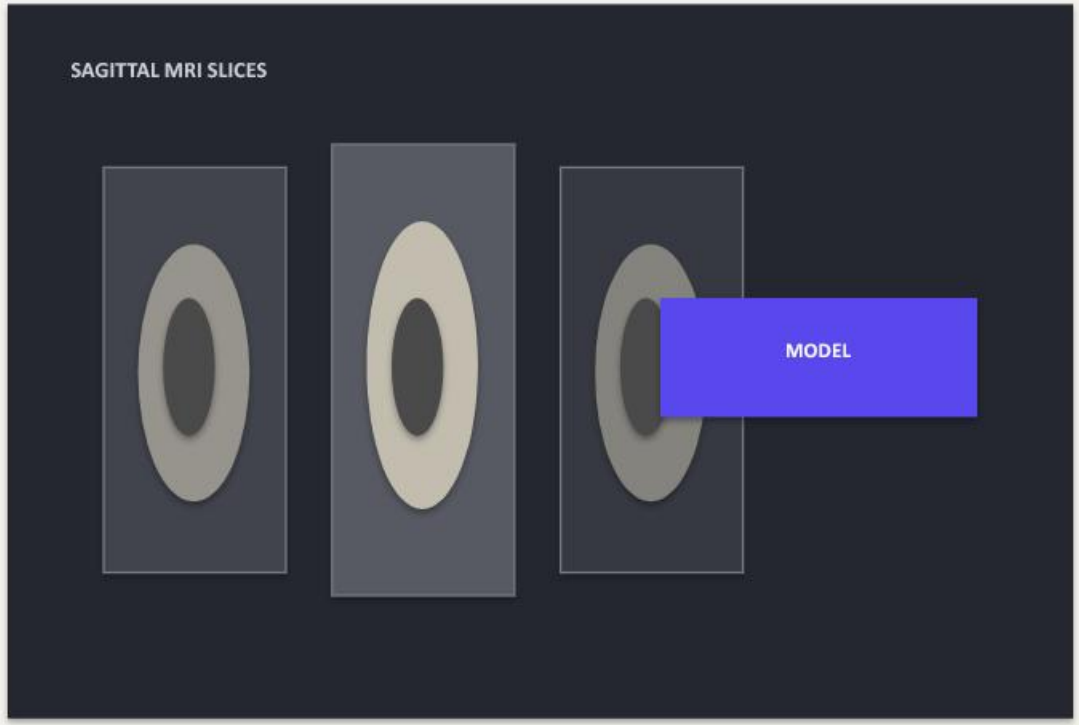
Classifying an ACL tear from a full MRI volume

A ResNet18 transfer-learning model reads sagittal knee MRI slices. The Streamlit demo accepts NumPy volumes, with optional DICOM and NIfTI readers.

0.9719

reported AUC-ROC

120-exam validation set / Stanford MRNet: 1,370 exams



Schematic only. Research prototype, not a diagnostic tool. The reported result has not been independently reproduced here.

Does the product photo match its description?

I used downloaded Rakuten France listings to compare text-only, image-only and fused models. The task pairs real listings with deliberately swapped images to create mismatch examples.

20,000

product listings in the subset

25%

images swapped for proxy labels



The mismatch label is a proxy signal, not a general moderation or policy decision.

Seasonal inputs lowered forecast error on the holdout

The research notebook uses Sentinel-2 NDVI features for Omo Forest Reserve and compares an LSTM with seasonal features against an NDVI-only baseline.

MODEL NOTEBOOK

The hosted dashboard uses illustrative values. The notebook contains the model and reported evaluation.



Small holdout with interpolated months. NDVI change alone does not establish deforestation.

Fine-tuning Gemma 2 for an African folktale challenge

I led Team Ethopie, a nine-person group, through the TRI AI Saturdays Cohort 10 capstone. I co-developed the Gemma 2 2B LoRA notebook with one teammate.

3rd

place in the cohort capstone

24 synthetic folktales / 38 prompt-story pairs



Synthetic benchmark. The score does not assess story quality or cultural fidelity.

Mapping surface water with Sentinel-2 indices

The Python project derives two water indices from Sentinel-2 bands, then fits a decision tree against historical surface-water labels. Spatial blocks separate evaluation areas.

Included

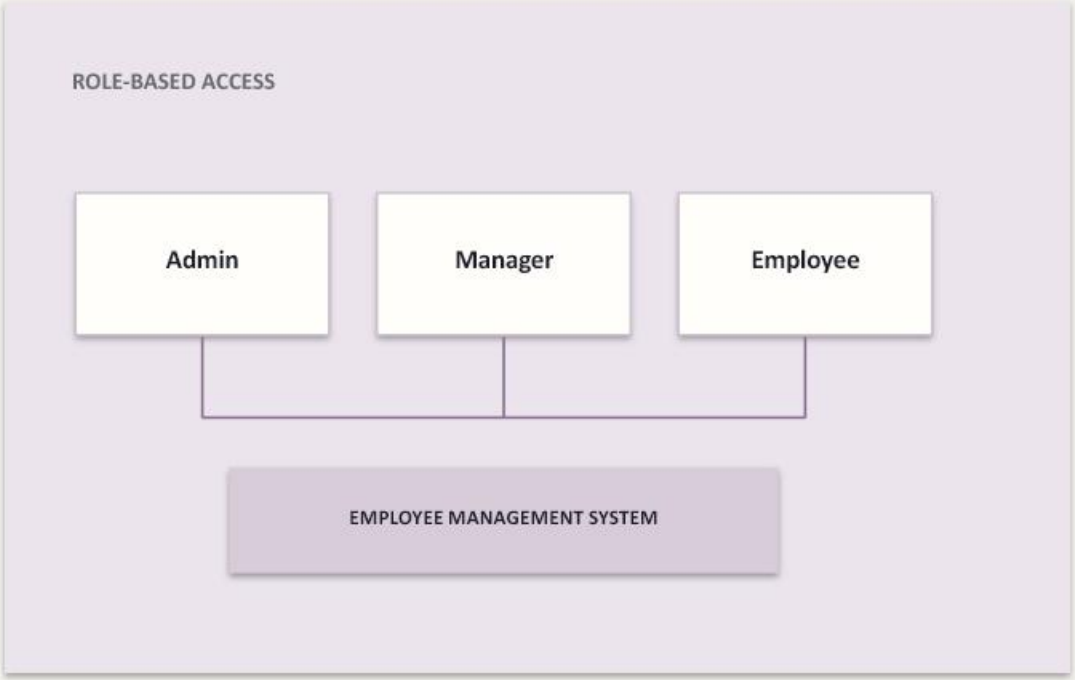
Threshold baselines
Documented map classes
Optional Earth Engine export



Generated demo data checks the pipeline only. Accuracy on real imagery still needs independent reference checks.

Employee records, attendance and payroll in one PHP app

Shebamiles EMS covers employee records, departments, attendance, leave, payroll and performance reviews. Each account has an admin, manager or employee role.



Security work

Prepared SQL statements
Password hashing and CSRF checks
First-admin setup through a CLI command

PHP checks and auth tests pass. Live MySQL integration has not been run in this workspace.

THANK YOU

Get in touch.

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Code, demos and project notes are linked from the portfolio.

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